

Task 1: Comparison Table

Feature	Amazon EBS	Amazon S3
Storage type	Block storage	Object storage
Attached to	EC2 instances	Accessible through APIs/console
Main use	OS, databases, applications	Images, documents, backups
Scalability	Volume-based	Highly scalable object storage
Access	Filesystem/block device	Object/key-based
Example	EC2 database	Instagram images

Task 2: Create and Attach EBS Volume

Steps

1. Open EC2.
2. Launch an EC2 instance.
3. Select an eligible instance configuration according to your AWS account's current Free Tier eligibility.
4. Launch the instance.
5. Open EC2 → Elastic Block Store → Volumes.
6. Click Create volume.
7. Select a suitable volume type such as gp3.
8. Set the size to at least 2 GB.
9. Make sure the volume is created in the same Availability Zone as the EC2 instance.
10. Create the volume.
11. Select the volume.
12. Choose Actions → Attach volume.

13. Select your EC2 instance.
14. Attach it.

Task 3: Format and Mount EBS

SSH into the EC2 instance.

First identify the disks:

- `lsblk`

Example:

- NAME SIZE TYPE MOUNTPOINT
- `nvme0n1` 8G disk
- `|—nvme0n1p1` 8G part /
- `nvme1n1` 2G disk

Identify the newly attached volume carefully.

Warning: Do not format the existing root disk. Formatting the wrong device can destroy the operating system/data.

Format the new volume:

- `sudo mkfs -t ext4 /dev/nvme1n1`

Create the mount directory:

- `sudo mkdir /mnt/mydata`

Mount it:

- `sudo mount /dev/nvme1n1 /mnt/mydata`

Verify:

- `df -h`

You can also run:

- `lsblk`

Task 4: Zomato-Style File Upload

Create the file:

- `echo "Zomato Menu - Pizza, Burger, Biryani" > zomato_menu.txt`

Copy it to the EBS volume:

- `sudo cp zomato_menu.txt /mnt/mydata/`

Verify:

- `ls -l /mnt/mydata/`

Display the file:

- `cat /mnt/mydata/zomato_menu.txt`

Task 5: Detach and Reattach EBS

Steps

1. Stop applications using the volume.
2. Unmount it:
 - `sudo umount /mnt/mydata`
3. In AWS Console, open EC2 → Volumes.
4. Select the EBS volume.
5. Choose Actions → Detach volume.
6. Wait until the volume becomes available.
7. Attach it to the same or another compatible EC2 instance.
8. On the new instance, identify the device:
 - `lsblk`
9. Mount it again:
 - `sudo mkdir /mnt/mydata`
 - `sudo mount /dev/nvme1n1 /mnt/mydata`
10. Check the stored file:
 - `ls -l /mnt/mydata/`